July 9th, 2023

Enhancing Retrieval-Augmented Large Language Models with Iterative Retrieval-Generation Synergy

Zhihong Shao, Yeyun Gong, Yelong Shen, Minlie Huang, Nan Duan, Weizhu Chen

Tsinghua University

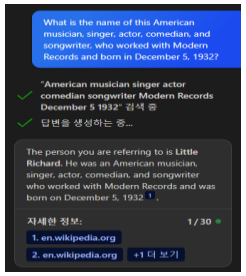
Microsoft Research Asia Microsoft

arXiv preprint arXiv:2305.15294 (2023)

발표자 : **박산희**
자연어 처리팀 : 황지현, 허정원, 이기성, 조해창

Introduction

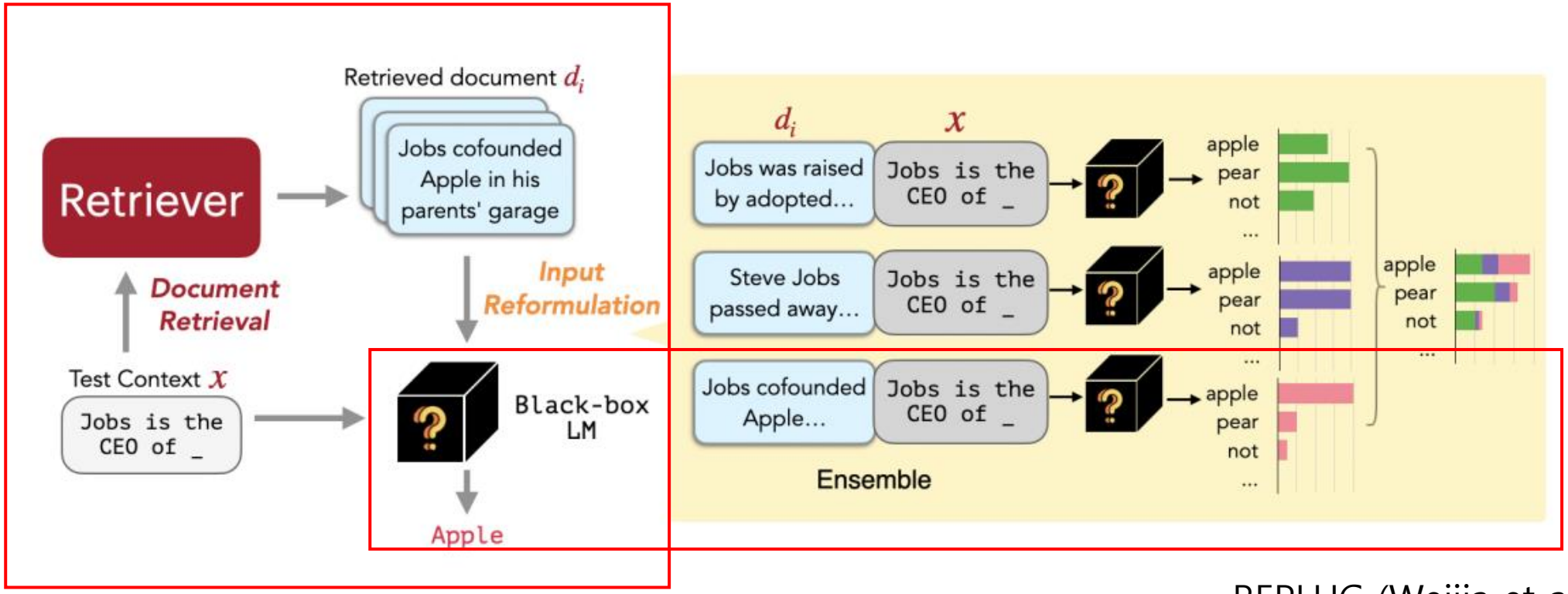
- LLM as text processor and reasoners
  - ChatGPT, Bing Chat …



- Retrieval-augmented language models
  - **LLMs to retrieve external knowledge.**
  - Conditioning LMs on retrieved documents
    - REALM (Guu et al., 2020), RAG (Lewis et al., 2020) , RETRO (Borgeaud et al., 2022), Atlas (Izacard et al., 2022)
  - Chain-of-thought prompting and retrievals
    - REPLUG (Weijia et al., 2023), *Iter-RetGen (Mialon, 2023)*

Mialon, Grégoire, et al. "Augmented language models: a survey." *arXiv preprint arXiv:2302.07842* (2023).

Introduction



REPLUG (Weijia et al., 2023)

Shi, Weijia, et al. "Replug: Retrieval-augmented black-box language models." arXiv preprint arXiv:2301.12652 (2023).

Introduction

- Retrieval-augmented LMs
  - One-time retrieval vs Multi-time retrieval
    - Factoid QA (e.g., Natural QA, single-hop fact verification…)

→ Retrieve answer based on stated original input

Example 1
Question: what color was john wilkes booth's hair
Wikipedia Page: John_Wilkes_Booth
Long answer: Some critics called Booth "the handsomest man in America" and a "natural genius", and noted his having an "astonishing memory"; others were mixed in their estimation of his acting. He stood 5 feet 8 inches (1.73 m) tall, had jet-black hair , and was lean and athletic. Noted Civil War reporter George Alfred Townsend described him as a "muscular, perfect man" with "curling hair, like a Corinthian capital".

Short answer: jet-black

- Multi-hop reasoning (e.g., Hotpot QA)

→ Retrieve multiple facts to obtain the answer

2

What is the name of this American musician, singer, actor, comedian, and songwriter, who worked with Modern Records and born in December 5, 1932?

1

Introduction

- Multi-hop reasoning, long-form question answering
  →Complex information needs

- How to solve them?
  → Multi-time retrieval
  - Gather required knowledge **multiple times through the generation process** (Trivedi et al., 2022)
  - **Search queries** as forward-looking sentences (Jiang et al., 2023)

Trivedi, Harsh, et al. "Interleaving retrieval with chain-of-thought reasoning for knowledge-intensive multi-step questions." arXiv preprint arXiv:2212.10509 (2022).

Introduction

- Previous works
  - GAR (Mao et al., 2021)
    - Generation query context with relevant passages (generation-augmented query → retrieval)
  - ReAct (Shunyu Yao, 2023)
    - Action space (Wikipedia web API) ←→ Reasoning (CoT with GPT3 or PaLM-540B)
  - Self-ask (Ofir, 2023)
    - Follow-up question with intermediate answer (search engine – google api) to ask before answering the final answer

- Limitation
  - During the generation process, some retrieved knowledge are failed.
  - Updated prompts by newly retrieved knowledge increase the overheads of retrieval and generation.

→ Need to enhance **Simple and Effective retrieval-augmented LLMs**

Mao, Yuning, et al. "Generation-augmented retrieval for open-domain question answering." *arXiv preprint arXiv:2009.08553* (2020).
Yao, Shunyu, et al. "ReAct: Synergizing Reasoning and Acting in Language Models." *The Eleventh International Conference on Learning Representations*. 2023.
Press, Ofir, et al. "Measuring and narrowing the compositionality gap in language models." *arXiv preprint arXiv:2210.03350* (2022).

Introduction

ReAct (Shunyu Yao, 2023)

(1) Hotspot QA

Question: Aside from the Apple Remote, what other device can control the program Apple Remote was originally designed to interact with?

ReAct

Thought 1: I need to search Apple Remote and find the program it was originally designed to interact with.
Act 1: Search[Apple Remote]
Obs 1: The Apple Remote is a remote control introduced in October 2005 by Apple … originally designed to control the Front Row media center program …

Thought 2: Apple Remote was originally designed to control the Front Row media center program. I need to search Front Row next and find what other device can control it.
Act 2: Search[Front Row]
Obs 2: Could not find [Front Row]. Similar: ['Front Row Seat to Earth', 'Front Row Motorsports','Front Row (software)', …]

Thought 3: Front Row is not found. I need to search Front Row (software) .
Act 3: Search[Front Row (software)]
Obs 3: Front Row is a discontinued media center software …

Thought 4: Front Row (software) is controlled by an Apple Remote or the keyboard function keys. So the answer is keyboard function keys.
Act 4: Finish[keyboard function keys]

Self-ask (Ofir, 2023)

Self-Ask

GPT-3

Question: Who lived longer, Theodor Haecker or Harry Vaughan Watkins?
Are follow up questions needed here: Yes.
Follow up: How old was Theodor Haecker when he died?
Intermediate answer: Theodor Haecker was 65 years old when he died.
Follow up: How old was Harry Vaughan Watkins when he died?
Intermediate answer: Harry Vaughan Watkins was 69 years old when he died.
So the final answer is: Harry Vaughan Watkins

Question: Who was president of the U.S. when superconductivity was discovered?
Are follow up questions needed here: Yes.
Follow up: When was superconductivity discovered?
Intermediate answer: Superconductivity was discovered in 1911.
Follow up: Who was president of the U.S. in 1911?
Intermediate answer: William Howard Taft.
So the final answer is: William Howard Taft.

Mao, Yuning, et al. "Generation-augmented retrieval for open-domain question answering." *arXiv preprint arXiv:2009.08553* (2020).
Yao, Shunyu, et al. "ReAct: Synergizing Reasoning and Acting in Language Models." *The Eleventh International Conference on Learning Representations*. 2023.
Press, Ofir, et al. "Measuring and narrowing the compositionality gap in language models." *arXiv preprint arXiv:2210.03350* (2022).

Iterative Retrieval-Generation Synergy

- **Overview**
  - Repeat the retrieval-generation for T iterations

Question: q

What is the height of the player who won the 2015 AFL Rising Star award?

Retrieval: $q \rightarrow \mathcal{D}_q$

Title: 2015 AFL Rising Star Context: The NAB AFL Rising Star award is given annually to a stand out young player in the Australian Football League. The award was won by Jesse Hogan of Melbourne

Retrieval-Augmented Generation: $\mathcal{D}_q \mid\mid q \rightarrow y_1$

The 2015 AFL Rising Star award was won by Jesse Hogan of Melbourne. Jesse Hogan is a professional Australian rules footballer. He is 198 cm tall. So the answer is 198 cm

Iteration 1

Question: q

What is the height of the player who won the 2015 AFL Rising Star award?

Generation-Augmented Retrieval: $y_1 \mid\mid q \rightarrow \mathcal{D}_{y_1 \mid\mid q}$

Title: Jesse Hogan Context: Jesse Hogan … playing for the Melbourne Football Club. A key forward, Hogan is 1.95 m tall … made his AFL debut in the 2015 season and won the Ron Evans Medal as the AFL Rising Star

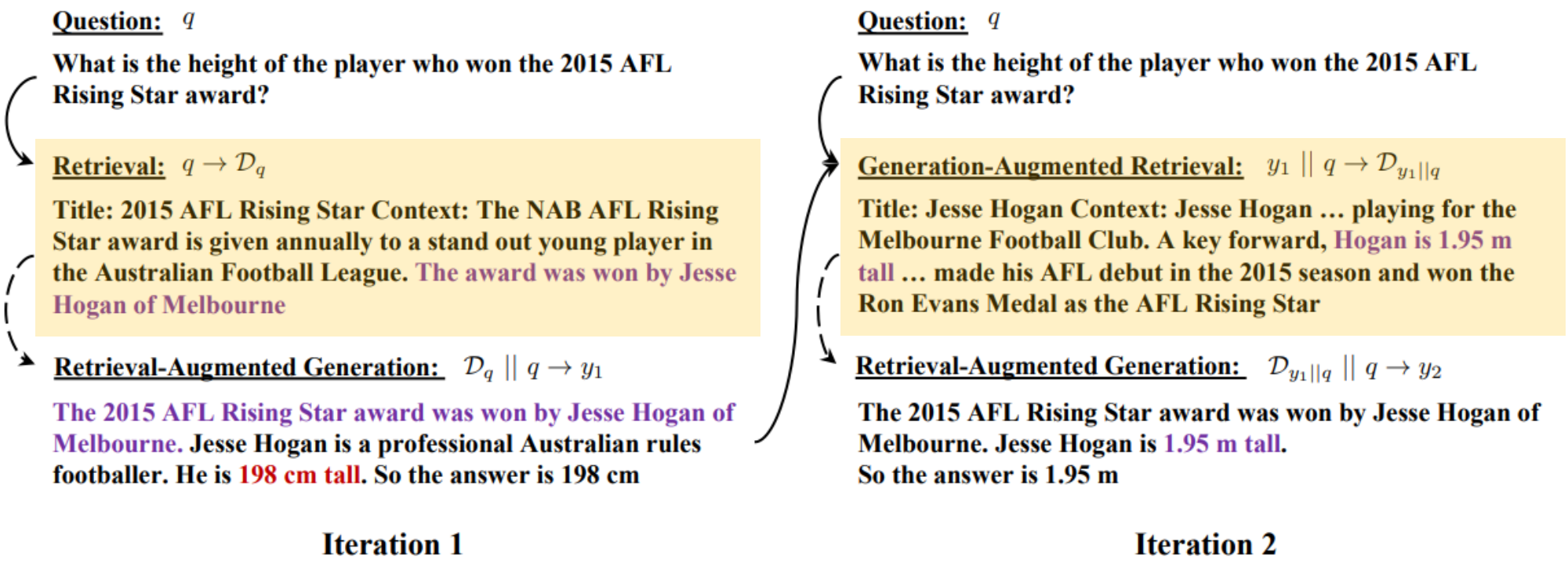
Retrieval-Augmented Generation: $\mathcal{D}_{y_1 \mid\mid q} \mid\mid q \rightarrow y_2$

The 2015 AFL Rising Star award was won by Jesse Hogan of Melbourne. Jesse Hogan is 1.95 m tall.
So the answer is 1.95 m

Iteration 2

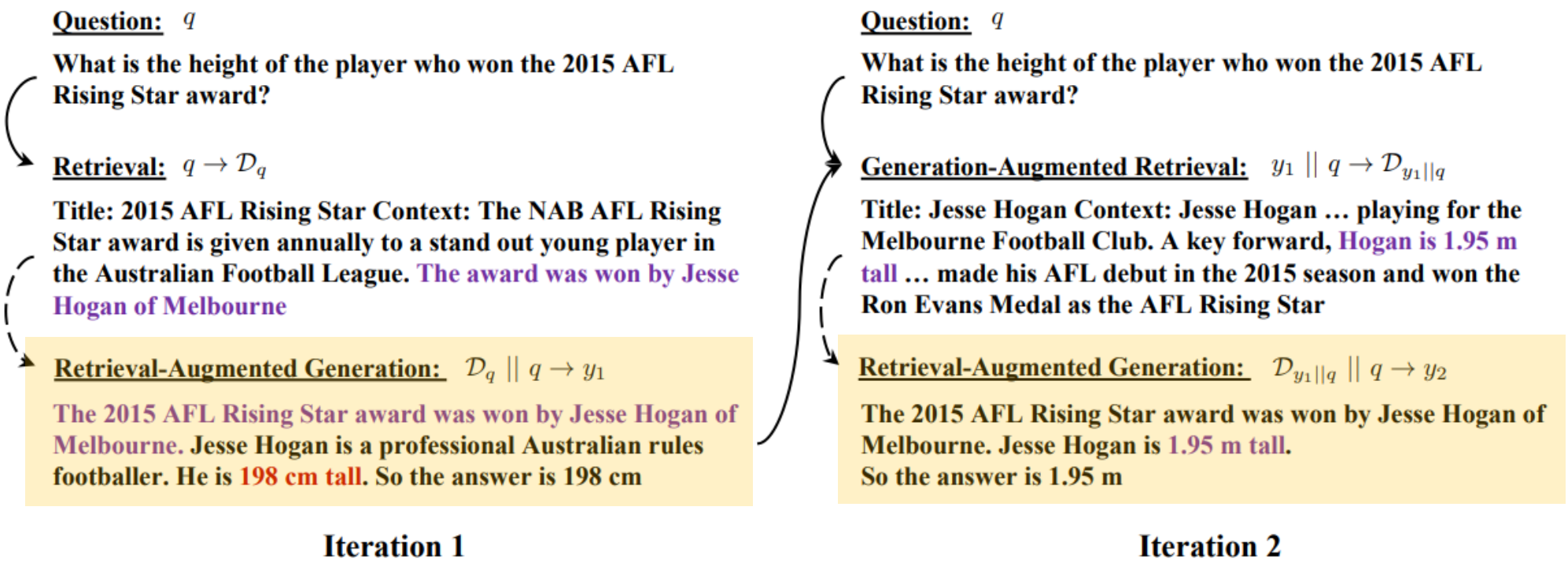
Iterative Retrieval-Generation Synergy

- **Retrieval Step**
  - Generation-augmented Retrieval
    - Iteration 1: $q \rightarrow D$
    - Iteration 2 – N: $y_{t-1}||q \rightarrow D$



Iterative Retrieval-Generation Synergy

- **Generation Step**
  - Retrieval-augmented Generation
    - Iteration 1: $D_q || q \rightarrow y_1$
    - Iteration 2 – N: $D_{y_{t-1||q}} || q \rightarrow y_t$



Iterative Retrieval-Generation Synergy

- **Generation Step**
  - Retrieval-augmented Generation
    - Iteration 1: $D_q || q \rightarrow y_1$
    - Iteration 2 – N: $D_{y_{t-1||q}} || q \rightarrow y_t$

LLM CoT style

$$y_t = M(y_t | prompt(D_{y_{t-1}||q}, \text{q}))$$

Question: q

What is the height of the player who won the 2015 AFL Rising Star award?

Retrieval: $q \rightarrow \mathcal{D}_q$

Title: 2015 AFL Rising Star Context: The NAB AFL Rising Star award is given annually to a stand out young player in the Australian Football League. The award was won by Jesse Hogan of Melbourne

Retrieval-Augmented Generation: $\mathcal{D}_q \,||\, q \rightarrow y_1$

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Generation-Augmented Retrieval: $y_1 \,||\, q \rightarrow \mathcal{D}_{y_1||q}$

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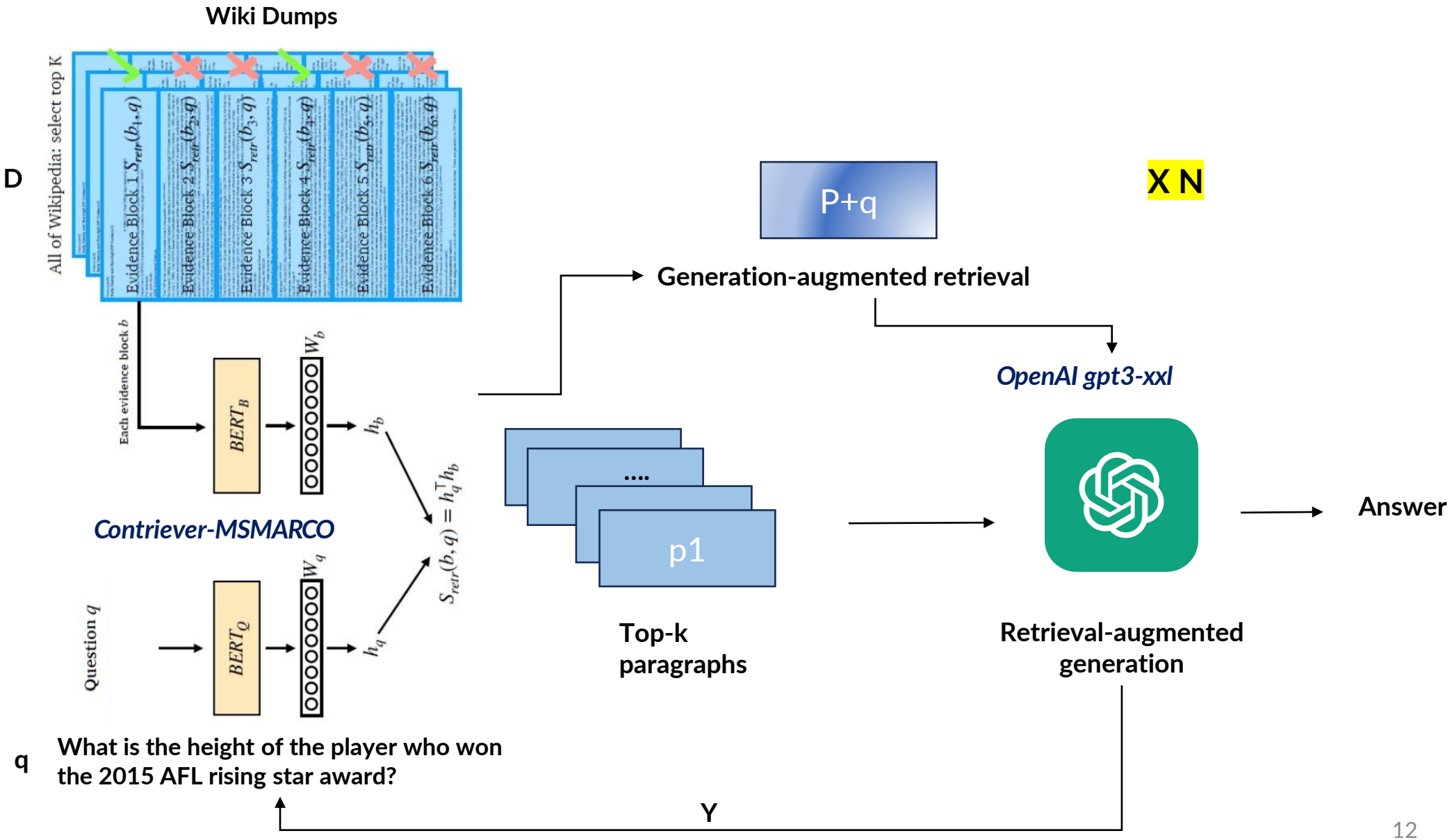
Retrieval-Augmented Generation: $\mathcal{D}_{y_1||q} \,||\, q \rightarrow y_2$

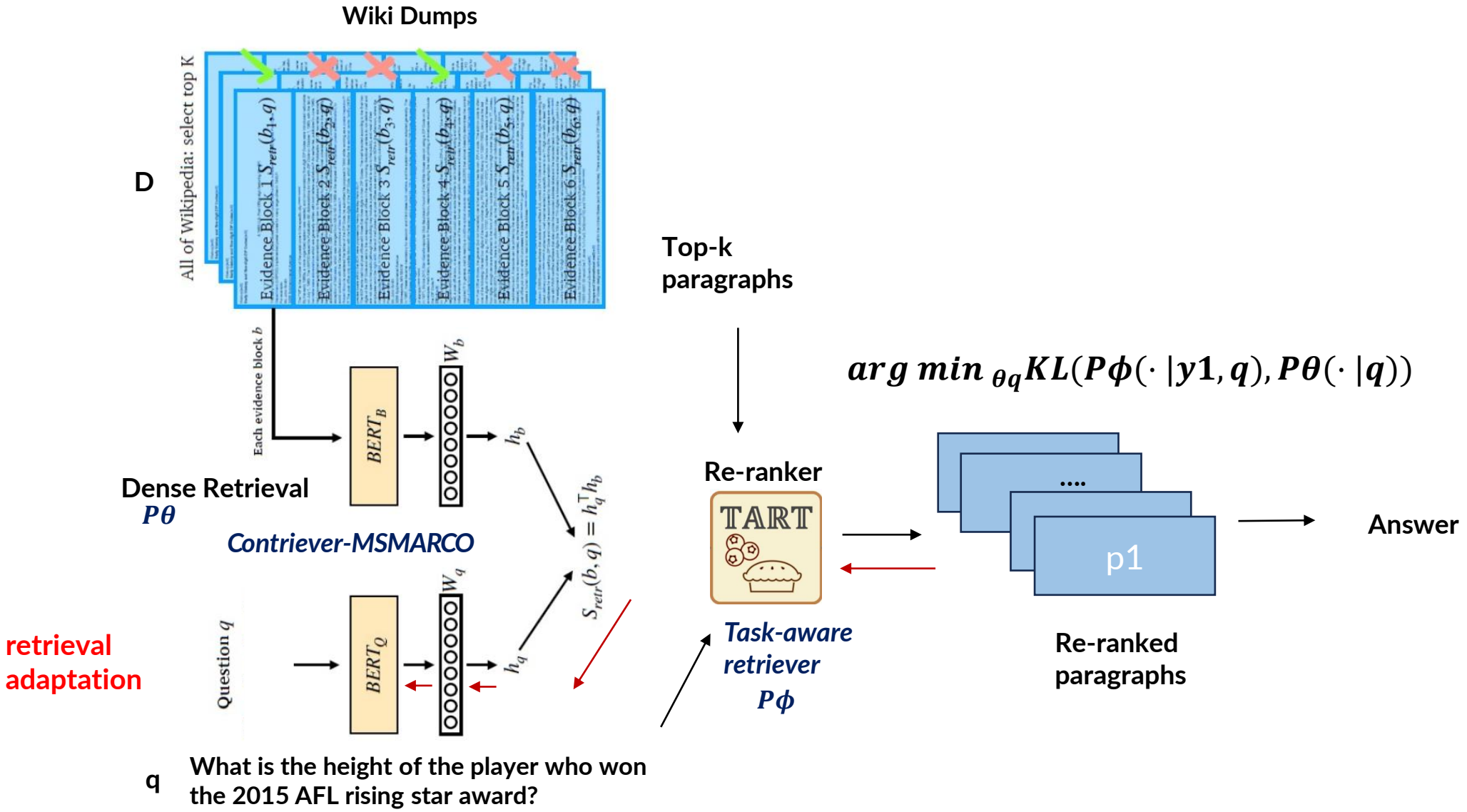
The 2015 AFL Rising Star award was won by Jesse Hogan of Melbourne. Jesse Hogan is 1.95 m tall.
So the answer is 1.95 m

Iteration 2

Flow


Wiki Dumps
D
All of Wikipedia: select top K
Evidence Block 1 $S_{retr}(b_1, q)$
Evidence Block 2 $S_{retr}(b_2, q)$
Evidence Block 3 $S_{retr}(b_3, q)$
Evidence Block 4 $S_{retr}(b_4, q)$
Evidence Block 5 $S_{retr}(b_5, q)$
Evidence Block 6 $S_{retr}(b_6, q)$
Each evidence block b
$BERT_B$
W_b
h_b
Contriever-MSMARCO
Question q
$BERT_Q$
W_q
h_q
$S_{retr}(b, q) = h_q^\top h_b$
q
What is the height of the player who won the 2015 AFL rising star award?
P+q
X N
Generation-augmented retrieval
OpenAI gpt3-xxl
....
p1
Top-k paragraphs
Retrieval-augmented generation
Answer
Y

Generation-augmented Retrieval Adaptation



Q & A

Experiments

- **Datasets**
  - Six datasets of three reasoning tasks

  - **Multi-hop question answering**
    - HotPotQA (Yang et al., 2018) , 2WikiMultiHopQA (Ho et al., 2020)
    - Wikipedia dump (Oct. 2017 ~ Dec. 2018, Dec. 2021) (Retrieved Documents)
    - MusiQue (Harsh Trivedi et al., 2021), Bamboogle (Ofir, 2023) – difficult answerable 2-hop 125 questions

  - **Fact verification**
    - Feverous (Alys et al., 2021)
    - Wikipedia dump (Dec. 2021)
  - **Commonsense reasoning**
    - StrategyQA (Geva et al., 2021)
    - Wikipedia dump (Dec. 2021)

facebookresearch/atlas: Code repository for supporting the paper "Atlas Few-shot Learning with Retrieval Augmented Language Models",(https//arxiv.org/abs/2208.03299) (github.com)
DPR/dpr/data/download_data.py at main · facebookresearch/DPR · GitHub

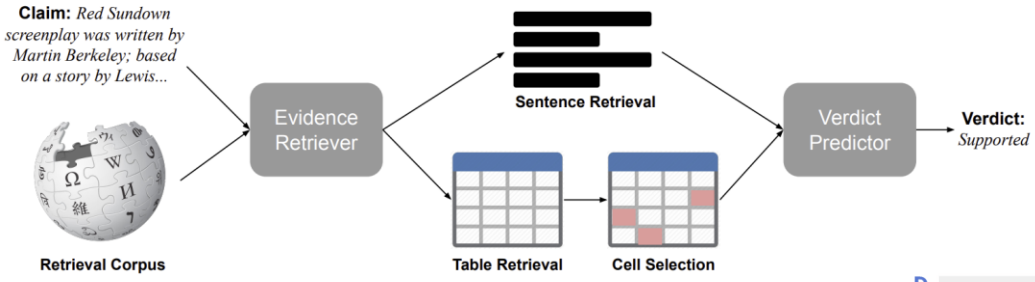
Dataset Details

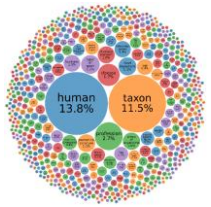
[Question]
Q: What city is the Marine Air Control Group 28 located in?
[Context]
P1: Marine Tactical Air Command Squadron 28 is a United States Marine Corps aviation command and control unit based at Marine Corps Air Station Cherry Point. They provide the 2nd Marine Aircraft Wings tactical headquarters and …
P2: Marine Corps Air Station Cherry Point or MCAS Cherry Point is a United States Marine Corps airfield located in Havelock, North Carolina, United States, in the eastern part of the state. It ...
P3-P10: …

2WikiMultiHopQA

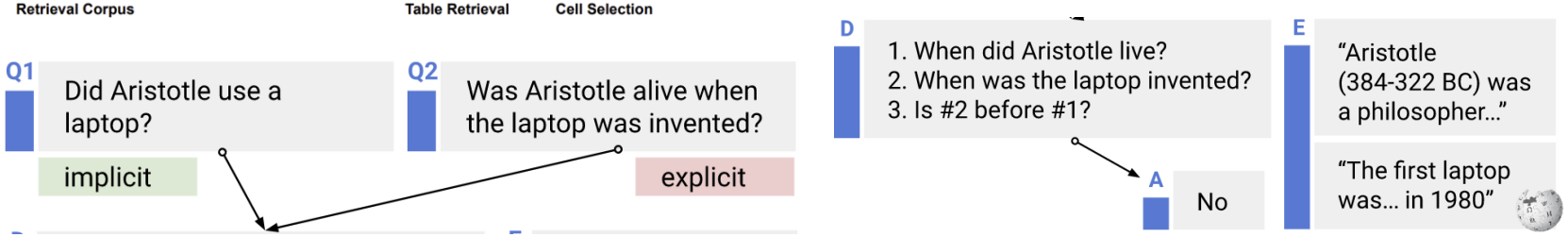
Reasoning Type	Example
Comparison question: comparing two entities	**Paragraph A:** Theodor Haecker (June 4, 1879 - April 9, 1945) was a … **Paragraph B:** Harry Vaughan Watkins (10 September 1875 – 16 May 1945) was a Welsh rugby union player … **Q:** Who lived longer, Theodor Haecker or Harry Vaughan Watkins?

FEVEROUS





StrategyQA



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DPR/dpr/data/download_data.py at main · facebookresearch/DPR · GitHub

Experiments

- **Evaluation Settings**
  - All methods are evaluated under the 3-shot setting
    - Sharing the same questions
  - Evaluation metrics : EM(exact match), F1, Acc‡
  - Evaluation template

Prompt for Evaluating the Correctness of a Model Output

In the following task, you are given a Question, a model Prediction for the Question, and a Ground-truth Answer to the Question. You should decide whether the model Prediction implies the Ground-truth Answer.

Question
{question}

Prediction
{model output}

Ground-truth Answer
{answer}

Does the Prediction imply the Ground-truth Answer? Output Yes or No:

Experiments

- **Baselines**
  - Direct Prompting
    - Q : The arena where the Lewiston Maineiacs played their home games can seat how many people? The answer is
  - CoT Prompting (thought-answer steps)
    - Q : What is the elevation range for the area that the eastern sector of the Colorado orogeny extends into? Let's think step by step. The eastern sector of Colorado orogeny extends into the High Plains. High Plains rise in elevation from around 1,800 to 7,000 ft, so the answer is 1,800 to 7,000 ft. The answer is
  - ReAct Prompting (thought-action-observation steps)
    - Q : What is the elevation range for the area that the eastern sector of the Colorado orogeny extends into?
    - Loop

      [**Search**/Lookup/Finish] The Colorado orogeny was an episode of mountain building (an orogeny) in Colorado and surrounding areas.

      The answer is
  - Self-Ask (Decomposing multi-hop question to sub-questions steps)
    - Q : When does monsoon season end in the state the area code 575 is located? Are follow up questions needed here: Yes. Follow up: Which state is the area code 575 located in? Intermediate answer: The area code 575 is located in New Mexico ... so the final answer is

Experiments

- **Results**

Multi-hop QA | Fact Verification | CQA

Method	HotPotQA			2WikiMultiHopQA			MuSiQue			Bamboogle			Feverous		StrategyQA	
	EM	F1	Acc†	EM	F1	Acc†	EM	F1	Acc†	EM	F1	Acc†	Acc	Acc†	Acc	Acc†
Without Retrieval																
Direct	21.9	36.8	44.8	21.3	29.2	33.9	7.0	18.7	15.8	11.2	24.4	28.0	60.1	60.1	66.5	66.7
CoT	30.0	44.1	50.0	30.0	39.6	44.0	19.4	30.9	28.6	**43.2**	**51.1**	60.0	59.8	59.8	71.0	71.0
With Retrieval																
Direct	31.6	44.7	53.3	27.3	35.4	43.6	13.9	28.2	26.5	17.6	31.8	43.2	69.8	69.8	65.6	65.6
ReAct	24.9	44.7	61.1	28.0	38.5	45.9	23.4	37.0	37.9	21.8	31.0	40.3	66.4	66.4	66.9	66.9
Self-Ask	36.8	55.2	64.8	**37.3**	**48.8**	55.9	**27.6**	41.5	**42.9**	31.5	41.2	54.8	70.7	70.7	70.2	70.2
ITER-RETGEN 1	39.2	53.9	65.5	33.7	45.2	55.4	24.2	38.6	38.1	36.8	47.7	57.6	67.0	67.0	72.0	72.0
ITER-RETGEN 2	44.1	58.6	71.2	34.9	47.0	58.1	26.4	41.1	41.0	38.4	48.7	59.2	68.8	68.8	73.0	73.0
ITER-RETGEN 3	45.2	59.9	71.4	34.8	47.8	58.3	25.7	41.4	40.8	37.6	47.0	59.2	69.0	69.0	72.3	72.3
ITER-RETGEN 4	45.8	**61.1**	**73.4**	36.0	47.4	58.5	26.7	41.8	40.8	38.4	49.6	60.0	**71.5**	**71.5**	73.8	73.8
ITER-RETGEN 5	45.2	60.3	72.8	35.5	47.5	58.8	25.7	40.7	39.6	39.2	49.7	**60.8**	70.3	70.3	73.2	73.2
ITER-RETGEN 6	**45.9**	61.0	73.3	35.5	48.1	**59.4**	25.9	40.5	39.8	40.0	50.0	59.2	70.9	70.9	72.4	72.4
ITER-RETGEN 7	45.1	60.4	72.9	35.5	47.4	58.4	26.1	**42.0**	41.0	40.0	50.7	**60.8**	70.5	70.5	**74.1**	**74.1**

Retrieval-augmented baselines

Experiments

- **Results**

Method	HotPotQA			2WikiMultiHopQA			MuSiQue			Bamboogle			Feverous		StrategyQA	
	EM	F1	Acc†	EM	F1	Acc†	EM	F1	Acc†	EM	F1	Acc†	Acc	Acc†	Acc	Acc†
Without Retrieval																
Direct	21.9	36.8	44.8	21.3	29.2	33.9	7.0	18.7	15.8	11.2	24.4	28.0	60.1	60.1	66.5	66.7
CoT	30.0	44.1	50.0	30.0	39.6	44.0	19.4	30.9	28.6	**43.2**	**51.1**	60.0	59.8	59.8	71.0	71.0
With Retrieval																
Direct	31.6	44.7	53.3	27.3	35.4	43.6	13.9	28.2	26.5	17.6	31.8	43.2	69.8	69.8	65.6	65.6
ReAct	24.9	44.7	61.1	28.0	38.5	45.9	23.4	37.0	37.9	21.8	31.0	40.3	66.4	66.4	66.9	66.9
Self-Ask	36.8	55.2	64.8	**37.3**	**48.8**	55.9	**27.6**	41.5	**42.9**	31.5	41.2	54.8	70.7	70.7	70.2	70.2
ITER-RETGEN 1	39.2	53.9	65.5	33.7	45.2	55.4	24.2	38.6	38.1	36.8	47.7	57.6	67.0	67.0	72.0	72.0
ITER-RETGEN 2	44.1	58.6	71.2	34.9	47.0	58.1	26.4	41.1	41.0	38.4	48.7	59.2	68.8	68.8	73.0	73.0
ITER-RETGEN 3	45.2	59.9	71.4	34.8	47.8	58.3	25.7	41.4	40.8	37.6	47.0	59.2	69.0	69.0	72.3	72.3
ITER-RETGEN 4	45.8	**61.1**	**73.4**	36.0	47.4	58.5	26.7	41.8	40.8	38.4	49.6	60.0	**71.5**	**71.5**	73.8	73.8
ITER-RETGEN 5	45.2	60.3	72.8	35.5	47.5	58.8	25.7	40.7	39.6	39.2	49.7	**60.8**	70.3	70.3	73.2	73.2
ITER-RETGEN 6	**45.9**	61.0	73.3	35.5	48.1	**59.4**	25.9	40.5	39.8	40.0	50.0	59.2	70.9	70.9	72.4	72.4
ITER-RETGEN 7	45.1	60.4	72.9	35.5	47.4	58.4	26.1	**42.0**	41.0	40.0	50.7	**60.8**	70.5	70.5	**74.1**	**74.1**

Experiments

- **Results**

Method	HotPotQA		2WikiMultiHopQA		MuSiQue		Bamboogle		Feverous		StrategyQA	
	# API	# Doc	# API	# Doc	# API	# Doc	# API	# Doc	# API	# Doc	# API	# Doc
ReAct	2.9	14.3	3.0	15.0	2.9	14.4	2.8	14.1	2.1	10.6	2.8	14.2
Self-Ask	3.2	16.0	3.2	15.9	3.0	14.8	3.0	14.9	2.3	11.3	3.0	15.1

Table 3: Average numbers of API calls to `text-davinci-003` and retrieved paragraphs for ReAct and Self-Ask. Note that ITER-RETGEN ($T = 2$) achieves significantly higher or competitive $\texttt{Acc}^{\dagger}$ with fewer API calls (i.e., 2) and fewer retrieved paragraphs (5 per iteration, 10 in total).

Dataset	HotPotQA			Feverous		
Retriever	Original	Distilled w/o y_1	Distilled w/ y_1	Original	Distilled w/o y_1	Distilled w/ y_1
ITER-RETGEN 1	65.5	67.1	**67.7**	67.0	67.3	**70.7**
ITER-RETGEN 2	71.2	75.2	**75.7**	68.8	68.1	**69.5**

Table 4: Effect of using LLM generation y_1 on optimizing a dense retriever. We evaluated ITER-RETGEN on HotPotQA and Feverous in terms of $\texttt{Acc}^{\dagger}$.

Experiments

- **Ablations**

Iteration	1	2	3	4	5	6	7
HotPotQA	49.5	66.1	65.7	66.5	66.7	66.7	67.1
2WikiMultiHopQA	29.0	45.2	46.2	46.7	45.8	45.8	46.5
MuSiQue	18.6	32.3	32.3	33.7	32.7	33.5	32.9
Bamboogle	20.8	36.0	36.8	36.0	35.2	36.0	36.0

Table 5: Answer recall of retrieved paragraphs at different iterations for ITER-RETGEN.

Subset	CoT ✓		CoT ✗		w/ Answer Retrieved		w/o Answer Retrieved	
Method	Self-Ask	ITER-RETGEN 2	Self-Ask	ITER-RETGEN 2	Self-Ask	ITER-RETGEN 2	Self-Ask	ITER-RETGEN 2
HotPotQA	77.5	**88.0**	52.0	**54.4**	78.1	**86.9**	29.9	**40.8**
2WikiMultiHopQA	68.8	**78.2**	**46.2**	42.0	73.1	**77.2**	30.1	**42.3**
MuSiQue	**68.5**	66.9	**32.6**	30.7	72.9	**78.9**	12.2	**22.9**
Bamboogle	73.0	**77.3**	28.0	**32.0**	76.2	**82.2**	32.8	**46.2**

- ITER-RETGEN consistently outperforms Self-Ask regardless of whether the in-context non-parametric knowledge mentions the answers or not.

Conclusion

- ITER-RETGEN Synergy is effective as answering questions with complex information.
- It outperforms retrieval-augmented baselines with simple workflow.
- ITER-RETGEN consistently leverage both parametric and non-parametric knowledge effectively.

Demos

- Test demo of ITER-RETGEN comparing with the self-ask method

```
ret = promptf(question_paper, prompt)

Question: Who lived longer, Muhammad Ali or Alan Turing?
Are follow up questions needed here: Yes.
Follow up: How old was Muhammad Ali when he died?
Intermediate answer: Muhammad Ali was 74 years old when he died.
Follow up: How old was Alan Turing when he died?
Intermediate answer: Alan Turing was 41 years old when he died.
So the final answer is: Muhammad Ali

Question: When was the founder of craigslist born?
Are follow up questions needed here: Yes.
Follow up: Who was the founder of craigslist?
Intermediate answer: Craigslist was founded by Craig Newmark.
Follow up: When was Craig Newmark born?
Intermediate answer: Craig Newmark was born on December 6, 1952.
So the final answer is: December 6, 1952

Question: Who was the maternal grandfather of George Washington?
Are follow up questions needed here: Yes.
Follow up: Who was the mother of George Washington?
Intermediate answer: The mother of George Washington was Mary Ball Washington.
Follow up: Who was the father of Mary Ball Washington?
Intermediate answer: The father of Mary Ball Washington was Joseph Ball.
So the final answer is: Joseph Ball

Question: Are both the directors of Jaws and Casino Royale from the same country?
...
Follow up: Who won the 2015 AFL Rising Star award?
Intermediate answer: Jesse Hogan.
Follow up: How tall is Jesse Hogan?
Intermediate answer: 6' 5".
So the final answer is: 6' 5"
```

Self-ASK

Direct Prompt

```
In [150]: prompt_test0 = question_paper + " " + "The answer is "

In [166]: prompt_test0
Out[166]: 'What is the height of the player who won the 2015 AFL Rising Star award? The answer is '

In [172]: print(bold("Direct prompt answer is"))
          print(yellowfy(llm(prompt_test0)))

Direct prompt answer is

Caleb Daniel, the winner of the 2015 AFL Rising Star award, is listed as being 5 feet 8 inches (173 cm) tall.
```

ITER-RETGEN

```
In [156]: prompt_test1 = p1 + " " + question_paper

Step 1 : Retrieval

In [180]: p1
Out[180]: 'Title:2015AFL Rising Star Context: The NAB AFL Rising Star awards is given annually to a stand out yonu player in the A
ustralian Football League. The award was won by Jesse Hogan of Melbourne.'

Step 2 : Retrieval-augmented Generation

In [179]: print(bold("Iter 1 : Paragraph 1 + question"))
          print()
          print(prompt_test1)

Iter 1 : Paragraph 1 + question

Title:2015AFL Rising Star Context: The NAB AFL Rising Star awards is given annually to a stand out yonu player in the Au
stralian Football League. The award was won by Jesse Hogan of Melbourne. What is the height of the player who won the 20
15 AFL Rising Star award?

In [174]: print(bold("Y1 is "))
          print(yellowfy(llm(prompt_test1)))

Y1 is

Jesse Hogan is 6 feet 4 inches (193 cm) tall.
```

ITER-RETGEN (zero-shot)

Demos

- Test demo of ITER-RETGEN comparing with the self-ask method

Step3 : Generation-augmented Retrieval

```
# paragraph 2 from y1 + q
p2 = "Title:Jesse Hogan Context:Jesse Hogan is playing for the Melbourne Football Club. A key forward, Jesse Hogan is
```

```
prompt_test2 = p2 + "₩n" + question_paper
```

```
# paragraph 2 from y1 + q
p2 = "Title:Jesse Hogan Context:Jesse Hogan is playing for the Melbourne Football Club. A key forward, Jesse Hogan is
```

```
print(bold("Paragraph 2 + question"))
print()
print(prompt_test2)
```

```
Paragraph 2 + question

Title:Jesse Hogan Context:Jesse Hogan is playing for the Melbourne Football Club. A key forward, Jesse Hogan is 6 feet 4
inches (193 cm) tall. his AFL debut in the 2015 season and wond the Ron Evans Medal as the AFL Rising Star
What is the height of the player who won the 2015 AFL Rising Star award?
```

```
print(bold("Y2 is "))
print(yellowfy(llm(prompt_test2)))
```

```
Y2 is

6 feet 4 inches (193 cm)
```

```
queries[5]
```

```
'2014 S/S is the debut album of a South Korean boy group that was formed by who?'
```

```
data_loader["validation"]["answer"][5]
```

```
'YG Entertainment'
```

```
passages[int(top_ids_and_scores[5][0][0])+1]
```

```
{'id': '3067812',
 'title': 'Japan in the late 1990s, the first-generation girl group S.E.S became the first Korean artists to debut in Japan in late 1998 and their first album
Reach Out in 1999. BoA₩'s debut Japanese album released in 2002, entitled "Listen to My Heart", was the first album by a Korean singer to debut at the top of the
Japanese Oricon Charts and become an RIAJ-certified "million-seller" in Japan. On January 16, 2008, TVXQ (known as "Tōhōshinki" in Japan) also reached the top of
the Oricon Charts with their sixteenth Japanese single "Purple Line". This made them the first Korean male group',
 'text': '3067813'}
```

→ Verification process is additionally required : checking the retrieval model performance.

- Retrieval test code with colab environment[1]

1 https://colab.research.google.com/drive/1UxmYCOh5jWXKjasmkNWoQ1BVGRzq_ul

Q & A

Thanks

Contact : carpediembackup20@gmail.com